Table of contents

Electro-Magnetic Field (force caused by potential)

- static: lovslized source charge at rest, only spacial dependency, no time dependency
- stationary: source charce at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistc: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movemnt
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m = \frac{1}{c}\left(\frac{r}{|r|} \times \mathcal{E}\right) \overset{\text{Far Field}}{=} \frac{1}{2}\frac{1}{2\pi}\mu_0 \frac{k^2 e^{i(kr-\omega t)}}{r}\frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_{0_{dipol}} \times \frac{r}{|r|}\right) \overset{\text{Near Field}}{\approx} \frac{1}{2}\frac{1}{2\pi}\mu_0 q \frac{v \times r}{|r|^2} \tag{1}$$

Magnetic Field in Matter

Exitation, Displacement

$$\mathcal{H} = ... \tag{2}$$

Electric Field in Vacuum

$$\mathcal{E} \overset{\text{conservativ}}{=} -\nabla U_e \overset{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \overset{\text{fast moving charge}}{=} \frac{1}{2}\frac{1}{2\pi}\frac{1}{\varepsilon_0}\frac{1}{r}\frac{1}{\left(1-\frac{r}{|r|}\cdot\frac{v}{c}\right)^3} q \left[\underbrace{\left(1-\frac{v^2}{c^2}\right)\frac{\frac{r}{|r|}-\frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right.\right.}_{\text{accelerate}} \tag{3}$$

Electric Field in Matter

Exitation, Displacement

$$\mathcal{D} = ... \tag{4}$$